

Paras Grammar School

Class 5

Chapter 1: Introduction to AI and its Applications

Answer the following questions.

a. What are the key stages involved in training an AI bot?

Following are the key stages involved in training an AI bot:

1. Collect Data
2. Prepare and Clean Data
3. Choose a Model
4. Train the Bot
5. Test and Evaluate
6. Improve and Retrain

b. What is the difference between two types of machine learning?

Supervised Learning	Unsupervised Learning
The models learns from labeled data	The model finds patterns in unlabeled data
Example: A computer learns to recognize cats and dogs using labeled data	Example: Grouping customers by shopping habits without knowing their preferences

c. How is AI beneficial in the field of healthcare? Provide one example.

AI can process large amounts of patient's data, including medical records and images. It assists doctors in making accurate diagnoses

identifying patterns and abnormalities the might be different to detect. AI is beneficial in medical imaging.

d. What is the key difference between machine learning and human learning?

Machine Learning	Human Learning
AI learns from lots of data and special rules to find patterns.	Human learn from what they observe in the real world.
Machines do not forget what they've learned.	Humans might forget things overtime, so we need to keep practicing and learning.

e. How does AI assist in the field of transportation, and what problems does it solve?

AI helps cars drive safely without a driver, reduces traffic jams by controlling signals, and suggests the fastest route with live traffic updates. It solves traffic congestion, prevents accidents by detecting danger and saves fuel and time.

f. How can AI benefit students in education?

AI helps students learn better and faster with smart tools.

1. Personalized Learning
2. Smart Tutors
3. Speech to Text
4. Language Translation
5. Automated Grading

g. Provide an example of an AI-powered learning app.

Duolingo is an AI-powered app that helps users learn new languages.

1. AI personalizes lessons based on progress.
2. Provides instant feedback on pronunciation and grammar.
3. Uses chatbots for practice conversations.

Chapter 2: Creating Media

Answer the following questions.

a. How can media files be imported into Clipchamp for a video product?

Click on the Your Media tab to view options for images, audio, and video. Import media to your template by selecting the required media and clicking on the + button.

b. What editing tools does Clipchamp offer to enhance a video?

Clipchamp makes video editing easy and fun. Following tools are used to enhance a video:

1. **Trim and Cut** – Remove unwanted part from video
2. **Transitions** – Smoothly switch between clips
3. **Filters and Effects** – Add color, slow motion and more
4. **Text and Titles** – Insert caption and stylish font
5. **Audio Editing** – Adjust volume, add background music
6. **Green Screen** – Replace backgrounds for cool effects

c. Is it possible to add text to a video in Clipchamp? If so, how?

Yes, you can easily add text in to a video in Clipchamp by following steps:

Step 1: Click *Text* from the toolbar

Step 2: Choose a style and drag it to the timeline

Step 3: Edit the text, font and size

d. Is it possible to add transitions to a video in Clipchamp? If so, how?

Yes, you can easily add transitions to a video in Clipchamp by following steps:

Step 1: Click *Transitions* from the toolbar

Step 2: Drag a transition between two clips

Step 3: Adjust the duration if needed

e. What is the process for exporting a finished video from Clipchamp?

Once your video is complete, save the project and export the video by following the steps:

Step 1: Click the Export button at the top-right

Step 2: Choose a resolution like 720p, 1080p or high quality

Step 3: Wait for the video to process

Step 4: Download the video to your device

f. What are the steps to add and customize text overlays in a video project using Microsoft Clipchamp?

We can add and customize text overlays in a video project by following the steps:

Step 1: Open Clipchamp and start a new project

Step 2: Click *Text* in the left toolbar

Step 3: Choose a text style and drag it onto the timeline

Step 4: Edit the text by typing your message

Step 5: Customize the font, size, color, and animation

Chapter 3: Fundamentals of Digital Marketing

Answer the following questions.

a. What is digital marketing in simple terms?

Digital marketing utilize the Internet to reach and engage customers. From online videos and TV ads to social media posts, digital marketing surrounds us daily. It is the activity of promoting and selling products or services which includes market research and advertising.

b. Why should businesses identify their Unique Selling Point (USP)?

Businesses should identify their Unique Selling Point (USP) because it features or benefits that sets a product, service, or business apart from competitors. For example, the USP for a handmade card business could be: ‘No two cards are the same, and we create them just for you to make every occasion extra special’.

c. What are the benefits of selling products online?

There are a lot of benefits of selling products online including:

1. **Wider Reach** – Sell to customers anywhere in the world
2. **Lower Costs** – No need for a physical store, saving money
3. **24/7 Availability** – Customers can shop anytime
4. **Easy Marketing** – Use social media and ads to attract buyers
5. **Convenience** – Manage your store from anywhere

d. How can businesses leverage search engines in their digital marketing strategy?

Potential clients can find your business through search engines like Google or Safari. They use different keywords to find products. If your website is optimized for popular keywords, it will rank higher.

e. How do social media interactions help promote online businesses?

Social media offers various ways to connect with others, like commenting, liking, or sharing posts. These interactions help build relationships and show your interests. Social media platforms use your activity to suggest relevant content and products.

f. Why is it crucial for businesses to understand their target audience?

It is important to understand the target audience due to various factors:

1. **Better Marketing** – Helps create ads that attract customers
2. **Increased Sales** – Products match customer needs, leading to more purchases
3. **Stronger Customer Loyalty** – Builds trust and long-term relationships
4. **Cost Efficiency** – Avoids wasting money on the wrong audience
5. **Improved Products** – Businesses can design products people actually want

Chapter 4: Building a Website

Answer the following questions.

a. Name the different types of HTML editors and give one example of each type.

There are different types HTML editors which helps in creating websites including:

1. **Text Editors** – Simple tools for writing HTML code. Example: Notepad
2. **WYSIWYG Editors** – “What you see is what you get” editors for visual design. Example: Adobe Dreamweaver
3. **Online Editors** – Web-based tools without installation. Example: JSFiddle

b. What are HTML elements?

A webpage is made up of HTML elements, which are identified by HTML tags. They are building blocks of a webpage. They consist of a start tag, content and an end tag.

c. Differentiate between container and empty elements.

Container	Empty Element
Container include opening and closing tag with content inside.	Empty Element have only an opening tag, no closing tag and no content.
Example: This is a paragraph	Hello World

d. List the various background properties in HTML.

There are various properties in HTML for background including:

1. ***background-color*** – Sets the background color
2. ***background-image*** – Adds an image as background
3. ***background-size*** – Defines the size of the background
4. ***background-repeat*** – Controls how the background repeats

5. ***background-position*** – Sets the position of the background image

6. ***background-attachment*** – Determines if background scrolls with the page

e. Look at your school's website and identify the background properties used on three different pages.

I have seen website of my school and it contains the following background properties:

1. background-color
2. background-image
3. background-repeat
4. background-position

f. Write an HTML code that displays your first name, age, sibling's first name, and your school's name.

```
<h1>About Me</h1>
<p><strong>My Name:</strong> John</p>
<p><strong>My Age:</strong> 10</p>
<p><strong>My Sibling's Name:</strong>
Emma</p>
<p><strong>My School's Name:</strong>
Paras Grammar School</p>
```

g. What is the purpose of Google Sites?

Google Sites is a free website-building tool by Google. It helps users create and share websites easily without coding. It is used for portfolios, business sites, and school projects.

Chapter 5: Data Handling with Google Sheets

Answer the following questions.

a. Write steps to create a customer template in Google Sheets that can be reused for educational purposes.

We can create a customer template in Google Sheets by following steps:

1. Open Google Sheets
2. Create a new Sheet
3. Design the template
4. Format the Sheet
5. Save as a Template
6. Move to Template Folder
7. Make a copy for reuse

b. What are the steps to format text within a cell in Google Sheets, including changing the font style and size?

We can format text within a cell in Google Sheets by following steps:

1. Select a Cell
2. Change font style from top menu
3. Change font size from top menu
4. Change the text color
5. Adjust Alignment

c. What are the steps to distribute Google Form via email to a specific group of recipients?

Click Share to enter email addresses or copy the sharing link to send the document to friends and family.

d. What are the steps to customize a Google Form?

We can customize the Google Form by following steps:

1. In Google Sheets, go to *Tools* and select *Create a new form*
2. Click *Add questions* here to start building your survey

3. Click the + button to add more questions
4. Enter the email addresses of your friends and family in the Send section and click on *Send*

e. List the steps for renaming a Google Sheet.

We can rename a Google Sheet by following steps:

1. Click on the title bar where name of the file is written
2. Double click on file name
3. Enter the new name you want to assign
4. Press *Enter*

f. Why is renaming important for file management?

Renaming is important in file management due to various reasons:

1. Easy identification
2. Better Organization
3. Efficient Searching
4. Avoids Duplication
5. Ensures Professionalism

Chapter 6: Coding Animal Adaptations

Answer the following questions.

- a. How can computer science and coding be used to model and understand animal adaptations? Provide an example.**

Coding helps create virtual models of animal behaviors and adaptations. AI and machine learning analyze animal movement, climate impact, and evolution. For example, Scientists use coding to simulate how polar bear adapts to climate change by tracking movement and hunting patterns in changing ice conditions.

- b. Explain the concept of events in programming.**

Events are actions or occurrences that triggers a response in a program. For example, in Sprite Lab, a block like “When sprite clicked” makes a character jump when clicked.

- c. How are events used in Sprite Lab to model animal adaptations?**

Events in Sprite Lab help simulate how animals react to their environment. They trigger actions like movement on conditions.

Examples:

1. **“When space key pressed”** – A chameleon changes color to blend into its surroundings.
2. **“When sprite touches water”** – A fish swims faster to escape predators.

- d. Describe a real-life example of an animal adaptation, and then explain how you would model it using Sprite Lab.**

Arctic foxes change their fur color with the seasons like white in winter for camouflage, brown in summer. We can model this animal adaptation in Sprite Lab by following steps:

1. Create a fox sprite
2. Use an event block **“when temperature changes”**

3. Use a costume change block

e. What is the significance of behaviors in animal survival, and how these behaviors be modeled in Sprite Lab?

Survival behaviors like hunting, escaping predators, and migration help animals adapt to their environment. We can model these survival behavior using Sprite Lab by using different events and sprites.

1. **Hunting** – “When a prey”, a lion sprite moves toward and catches the prey
2. **Escape** – “When enemy touches”, a rabbit sprite jumps away to avoid predators
3. **Migration** – “When winter starts,” birds move to a warmer location